Moment Generating Functions.

For a r.v. Y,
and for an independent argument denoted by t,
we define the moment generating function (mgf) of Y as this function of t:

$$M_Y(t) := \mathbb{E}\left[e^{Y \cdot t}\right]$$ for all t such that the expectation exists

Q: $M_Y(0) = \underline{1}$ $\Rightarrow$ @ least $t=0$ is in the domain

We say that the mgf exists if it is finite for t such that $|t| \leq b$ for some $b>0$. $\nu = nu$

<u>Goal</u>: To understand e^X w/ $X \sim \text{Normal}(\text{mean} = m, \text{var} = \nu^2)$

<u>Recall</u>: In terms of $Z \sim N(0,1)$,

$$\boxed{X = m + \nu \cdot Z}$$

<u>Fact</u>: $\boxed{M_Z(t) = e^{t^2/2}}$ for all $t \in \mathbb{R}$

$\Rightarrow$ For any normal X:

$$M_X(t) = \mathbb{E}\left[e^{X \cdot t}\right] = \mathbb{E}\left[e^{(m+\nu Z)\cdot t}\right]$$
$$= \mathbb{E}\left[e^{mt} \cdot e^{\nu t \cdot Z}\right].$$
$$= e^{mt}\, \mathbb{E}\left[e^{\nu t \cdot Z}\right]$$
$$= e^{mt} M_Z(\nu t) = e^{mt} \cdot e^{\frac{\nu^2 t^2}{2}} = \boxed{e^{mt + \frac{\nu^2 t^2}{2}}}$$

The lognormal distribution.

Definition 1.1. Let $X \sim Normal(mean = m, variance = \nu^2)$. Define the random variable $Y = e^X$. We say that the random variable Y is *lognormally distributed*.

1.1. First properties.

- The expected value of the lognormally distributed random variable Y can be obtained as follows:

$\mathbb{E}[e^{1\cdot X}]$

$$\mathbb{E}[Y] = \mathbb{E}[e^X] = M_X(1) = \boxed{e^{m+\frac{\nu^2}{2}}}.$$

$= e^{aX}$

- Let Y be a lognormal and let $a \neq 0$. Then, the random variable Y^a is also lognormal. *Note:* For $a = 0$, we get a degenerate random variable at 1 which can, technically, be interpreted as lognormal, but is not fun.
- Let Y_1 and Y_2 be independent and lognormally distributed. Then, Y_1Y_2 is also lognormal.

1.2. Quantiles.

Definition 1.2. For p such that $0 < p < 1$, we define the $100p^{th}$ *quantile* of a random variable X as any value π_p such that

$$F_X(\pi_p-) \leq p \leq F_X(\pi_p).$$

In particular, the 50th quantile is referred to as the *median*.

Note: When the random variable X is continuous, we can obtain the $100p^{th}$ quantile by simply solving for π_p in

$$F_X(\pi_p) = p.$$

Consider a probability p. Let z_p be the $100p^{th}$ quantile of the standard normal distribution. Let Y be lognormally distributed as above. My claim is that the value

$$y_p = e^{m+\nu z_p}$$

is the $100p^{th}$ quantile of Y. Let us simply verify this claim by calculating $F_Y(y_p)$. We have, with $Z \sim N(0,1)$,

$$F_Y(y_p) = \mathbb{P}[Y \leq y_p] = \mathbb{P}[e^X \leq y_p] = \mathbb{P}[e^{m+\nu Z} \leq e^{m+\nu z_p}].$$

Since the logarithmic function is increasing, we have that the above equals

$$F_Y(y_p) = \mathbb{P}[m + \nu Z \leq m + \nu z_p] = \mathbb{P}[Z \leq z_p] = p.$$

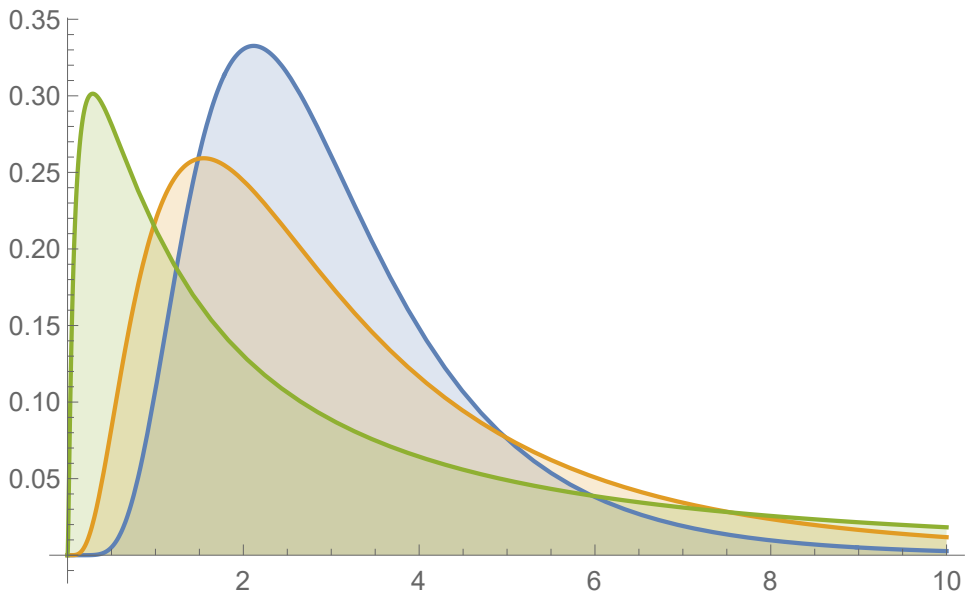
The above concludes our proof.

In particular, since the median of the standard normal distribution equals 0, the median of the lognormal distribution will be e^m.

Note: Since

$$\boxed{e^m < e^{m+\frac{\nu^2}{2}},} \tag{1.1}$$

i.e., since the mean of a lognormal distribution always exceeds the median, we say that it's *right-skewed*. In fact, this is what its probability density function looks like.


0.35
0.30
0.25
0.20
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0.10
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2
4
6
8
10

Jensen's Inequality.

Caveat:

$$\boxed{\mathbb{E}[e^X] \geq e^{\mathbb{E}[X]}}$$

$X \sim \text{Normal}(\text{mean}=m, \text{var}=\nu^2)$

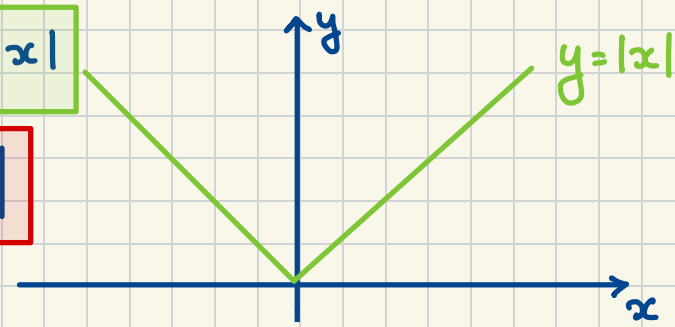
Theorem. Let X be a r.v., and let g be a convex function such that $g(X)$ is well defined and $\mathbb{E}[g(X)]$ exists.

Then,

$$\boxed{\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])}.$$

Example. i. $\boxed{g(x) = |x|}$

$$\boxed{\mathbb{E}[|X|] \geq |\mathbb{E}[X]|}$$

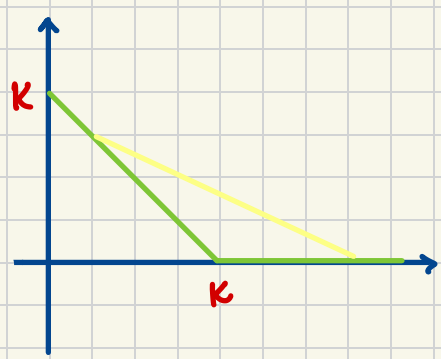


ii. Consider a European put w/ strike K.

Its payoff f'tion: $v_p(s) = (K-s)_+$

The expected payoff

$$\mathbb{E}[v_p(S(T))] = \mathbb{E}[(K-S(T))_+]$$

By Jensen's Inequality it's

$$\geq (K - \mathbb{E}[S(T)])_+$$